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LAB – 9

To implement BEFORE and AFTER triggers for INSERT, UPDATE and DELETE operations in MySQL.

- Create the database and tables

```
mysql> CREATE TABLE EmployeeAudit (
    ->     audit_id INT AUTO_INCREMENT PRIMARY KEY,
    ->     emp_id INT,
    ->     action_type VARCHAR(20),
    ->     action_time TIMESTAMP DEFAULT CURRENT_TIMESTAMP
    -> );
Query OK, 0 rows affected (0.01 sec)

mysql>
mysql> INSERT INTO Employee10 VALUES
    -> (1, 'Asha', 30000),
    -> (2, 'Rahul', 35000),
    -> (3, 'Priya', 40000);
Query OK, 3 rows affected (0.00 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql>
mysql> SELECT * FROM Employee10;
+--------+----------+----------+
| emp_id | emp_name | salary   |
+--------+----------+----------+
|      1 | Asha     | 30000.00 |
|      2 | Rahul    | 35000.00 |
|      3 | Priya    | 40000.00 |
+--------+----------+----------+
3 rows in set (0.00 sec)
```

- BEFORE INSERT trigger

```
mysql> CREATE TRIGGER before_employee_insert
    -> BEFORE INSERT ON Employee10
    -> FOR EACH ROW
    -> BEGIN
    ->     IF NEW.salary <= 0 THEN
    ->         SIGNAL SQLSTATE '45000'
    ->         SET MESSAGE_TEXT = 'Salary must be positive';
    ->     END IF;
    -> END //
Query OK, 0 rows affected (0.02 sec)

mysql>
mysql> DELIMITER ;
mysql> INSERT INTO Employee10
    -> VALUES (4, 'Kiran', 42000);
Query OK, 1 row affected (0.01 sec)

mysql>
mysql> SELECT * FROM Employee10;
+--------+----------+----------+
| emp_id | emp_name | salary   |
+--------+----------+----------+
|      1 | Asha     | 30000.00 |
|      2 | Rahul    | 35000.00 |
|      3 | Priya    | 40000.00 |
|      4 | Kiran    | 42000.00 |
+--------+----------+----------+
4 rows in set (0.00 sec)
```

- BEFORE UPDATE trigger

```
mysql> CREATE TRIGGER before_employee_update
    -> BEFORE UPDATE ON Employee10
    -> FOR EACH ROW
    -> BEGIN
    ->     IF NEW.salary <= 0 THEN
    ->         SIGNAL SQLSTATE '45000'
    ->         SET MESSAGE_TEXT = 'Salary must be positive';
    ->     END IF;
    -> END //
Query OK, 0 rows affected (0.01 sec)

mysql>
mysql> DELIMITER ;
mysql> UPDATE Employee10
    -> SET salary = 45000
    -> WHERE emp_id = 1;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql>
mysql> SELECT * FROM Employee10;
+--------+----------+----------+
| emp_id | emp_name | salary   |
+--------+----------+----------+
|      1 | Asha     | 45000.00 |
|      2 | Rahul    | 35000.00 |
|      3 | Priya    | 40000.00 |
|      4 | Kiran    | 42000.00 |
+--------+----------+----------+
```

- BEFORE DELETE trigger

```
mysql> CREATE TRIGGER before_employee_delete
    -> BEFORE DELETE ON Employee10
    -> FOR EACH ROW
    -> BEGIN
    ->     IF OLD.emp_id = 1 THEN
    ->         SIGNAL SQLSTATE '45000'
    ->         SET MESSAGE_TEXT = 'Cannot delete protected employee';
    ->     END IF;
    -> END //
Query OK, 0 rows affected (0.01 sec)

mysql>
mysql> DELIMITER ;
mysql> DELETE FROM Employee10
    -> WHERE emp_id = 2;
Query OK, 1 row affected (0.02 sec)

mysql>
mysql> SELECT * FROM Employee10;
+--------+----------+----------+
| emp_id | emp_name | salary   |
+--------+----------+----------+
|      1 | Asha     | 45000.00 |
|      3 | Priya    | 40000.00 |
|      4 | Kiran    | 42000.00 |
+--------+----------+----------+
3 rows in set (0.00 sec)
```

- AFTER INSERT trigger

```
mysql> CREATE TRIGGER after_employee_insert
    -> AFTER INSERT ON Employee10
    -> FOR EACH ROW
    -> BEGIN
    ->     INSERT INTO EmployeeAudit (emp_id, action_type)
    ->     VALUES (NEW.emp_id, 'INSERT');
    -> END //
Query OK, 0 rows affected (0.01 sec)

mysql>
mysql> DELIMITER ;
mysql> INSERT INTO Employee10
    -> VALUES (6, 'Meena', 32000);
Query OK, 1 row affected (0.01 sec)

mysql>
mysql> SELECT * FROM EmployeeAudit;
+----------+--------+-------------+---------------------+
| audit_id | emp_id | action_type | action_time         |
+----------+--------+-------------+---------------------+
|        1 |      6 | INSERT      | 2026-10-09 19:15:13 |
+----------+--------+-------------+---------------------+
1 row in set (0.00 sec)
```

- AFTER UPDATE trigger

```
mysql> CREATE TRIGGER after_employee_update
    -> AFTER UPDATE ON Employee10
    -> FOR EACH ROW
    -> BEGIN
    ->     INSERT INTO EmployeeAudit (emp_id, action_type)
    ->     VALUES (NEW.emp_id, 'UPDATE');
    -> END //
Query OK, 0 rows affected (0.01 sec)

mysql>
mysql> DELIMITER ;
mysql> UPDATE Employee10
    -> SET salary = 36000
    -> WHERE emp_id = 6;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql>
mysql> SELECT * FROM EmployeeAudit;
+----------+--------+-------------+---------------------+
| audit_id | emp_id | action_type | action_time         |
+----------+--------+-------------+---------------------+
|        1 |      6 | INSERT      | 2026-10-09 19:15:13 |
|        2 |      6 | UPDATE      | 2026-10-09 19:15:49 |
+----------+--------+-------------+---------------------+
2 rows in set (0.00 sec)
```

- AFTER DELETE trigger

```
mysql> DELIMITER //
mysql>
mysql> CREATE TRIGGER after_employee_delete
    -> AFTER DELETE ON Employee10
    -> FOR EACH ROW
    -> BEGIN
    ->     INSERT INTO EmployeeAudit (emp_id, action_type)
    ->     VALUES (OLD.emp_id, 'DELETE');
    -> END //
Query OK, 0 rows affected (0.01 sec)

mysql>
mysql> DELIMITER ;
mysql> DELETE FROM Employee10
    -> WHERE emp_id = 6;
Query OK, 1 row affected (0.01 sec)

mysql>
mysql> SELECT * FROM EmployeeAudit;
+----------+--------+-------------+---------------------+
| audit_id | emp_id | action_type | action_time         |
+----------+--------+-------------+---------------------+
|        1 |      6 | INSERT      | 2026-10-09 19:15:13 |
|        2 |      6 | UPDATE      | 2026-10-09 19:15:49 |
|        3 |      6 | DELETE      | 2026-10-09 19:16:41 |
+----------+--------+-------------+---------------------+
3 rows in set (0.00 sec)
```

- Display all audit records

```
mysql> SELECT * FROM EmployeeAudit
    -> ORDER BY audit_id;
+----------+--------+-------------+---------------------+
| audit_id | emp_id | action_type | action_time         |
+----------+--------+-------------+---------------------+
|        1 |      6 | INSERT      | 2026-10-09 19:15:13 |
|        2 |      6 | UPDATE      | 2026-10-09 19:15:49 |
|        3 |      6 | DELETE      | 2026-10-09 19:16:41 |
+----------+--------+-------------+---------------------+
3 rows in set (0.00 sec)
```